

A $0 < q \leq 5$ partial solution to the non-negative Hunter moment conjecture

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Abstract

Brazitikos and Pandis conjecture that, for non-negative coefficients $\sum_i a_i^2 = 1$, the real moment $\mathbb{E}(a_1 X_1 + \dots + a_n X_n)^q$ is bounded below by the least equal-support moment $\min_{1 \leq m \leq n} \rho(m, q)$. This packet proves the conjecture for all dimensions in the range $0 < q \leq 5$. The new step beyond the known $q \leq 4$ Schur-concavity range is to combine the source paper's integer endpoint $k = 5$ theorem with Borell log-concavity of normalized moments.

1 Source question

The source is:

Silouanos Brazitikos and Christos Pandis, *Sharp inequalities for symmetric polynomials, Hunter's conjecture, and moments of exponential random variables*, arXiv:2512.12254.

The relevant open problem is Conjecture 4.8 in the source PDF:

We also conjecture that this behavior holds for every real $q > 0$, as expressed in the following conjecture:

Conjecture 4.8. Let $q > 0$ be a real number, and let $a_1, \dots, a_n$ be non-negative real numbers satisfying $\sum_{i=1}^n a_i^2 = 1$. If $X_1, \dots, X_n$ are i.i.d. standard exponential random variables, then

$$\mathbb{E}(a_1 X_1 + \dots + a_n X_n)^q \geq \min\{\rho(1, q), \dots, \rho(n, q)\}.$$

4.1. Characterization of Extrema. We first introduce a Lemma that will be useful for the characterization of the global extrema

For $m \in \mathbb{N}$ and $q > 0$, write

$$\rho(m, q) = \mathbb{E} \left(\frac{X_1 + \dots + X_m}{\sqrt{m}} \right)^q = \frac{\Gamma(m+q)}{\Gamma(m) m^{q/2}},$$

where the X_i are i.i.d. standard exponential random variables.

2 Proof intuition

The source paper proves the conjectured lower bound at every non-negative integer exponent. At the integer $k = 5$, the least equal-support moment is still the one-coordinate value $\rho(1, 5) = \Gamma(6)$. On the other hand, any positive linear combination of independent exponentials has a log-concave density, and Borell's lemma says that

$$q \mapsto \frac{\mathbb{E}Y^q}{\Gamma(q+1)}$$

is log-concave. Interpolating between $q = 0$, where the normalized moment is 1, and $q = 5$, where the source's integer theorem gives a value at least 1, forces the normalized moment to stay at least 1 throughout $[0, 5]$. A separate scalar lemma checks that, throughout the same interval, $\rho(1, q)$ is exactly the minimum among the $\rho(m, q)$.

3 Scalar minimum through $q = 5$

Lemma 1. *For every integer $m \geq 1$ and every $0 \leq q \leq 5$,*

$$\rho(m, q) \geq \rho(1, q) = \Gamma(q+1).$$

Consequently, for every $n \geq 1$,

$$\min_{1 \leq m \leq n} \rho(m, q) = \rho(1, q), \quad 0 \leq q \leq 5.$$

Proof. For $m = 1$ there is equality. Fix $m \geq 2$ and set

$$R_m(q) = \frac{\rho(m, q)}{\Gamma(q+1)} = \frac{\Gamma(m+q)}{\Gamma(m)\Gamma(q+1)m^{q/2}}.$$

The function $\log R_m(q)$ is concave on $[0, \infty)$, because

$$\frac{d^2}{dq^2} \log R_m(q) = \psi_1(m+q) - \psi_1(q+1) \leq 0,$$

where ψ_1 is the trigamma function, which is decreasing on $(0, \infty)$. Since $R_m(0) = 1$, it remains enough to show $R_m(5) \geq 1$.

At $q = 5$,

$$R_m(5) = \frac{(m+1)(m+2)(m+3)(m+4)}{120m^{3/2}}.$$

For real $m \geq 2$, let the right-hand side be $r(m)$. Then

$$\frac{d}{dm} \log r(m) = \sum_{j=1}^4 \frac{1}{m+j} - \frac{3}{2m} = \frac{5m^4 + 30m^3 + 35m^2 - 50m - 72}{2m(m+1)(m+2)(m+3)(m+4)} > 0$$

for $m \geq 2$. Thus $r(m) \geq r(2) = 3/(2\sqrt{2}) > 1$ for all $m \geq 2$. By concavity of $\log R_m$,

$$\log R_m(q) \geq \left(1 - \frac{q}{5}\right) \log R_m(0) + \frac{q}{5} \log R_m(5) \geq 0$$

for $0 \leq q \leq 5$. This proves the lemma. □

4 Partial solution

Theorem 1. *Let $0 < q \leq 5$, let $a_1, \dots, a_n \geq 0$ satisfy $\sum_{i=1}^n a_i^2 = 1$, and let $X_1, \dots, X_n$ be i.i.d. standard exponential random variables. Then*

$$\mathbb{E}(a_1 X_1 + \dots + a_n X_n)^q \geq \min_{1 \leq m \leq n} \rho(m, q).$$

In fact, the stronger bound

$$\mathbb{E}(a_1 X_1 + \dots + a_n X_n)^q \geq \Gamma(q+1)$$

holds throughout this range.

Proof. Discard zero coefficients and put

$$Y = a_1 X_1 + \dots + a_n X_n.$$

For $a_i > 0$, the random variable $a_i X_i$ has log-concave density

$$t \mapsto a_i^{-1} e^{-t/a_i} \mathbf{1}_{[0, \infty)}(t).$$

By the Prekopa-Leindler theorem, convolution preserves log-concavity, so Y has a log-concave density on $(0, \infty)$.

The source paper's Lemma 4.13, a Borell log-concavity lemma, therefore applies:

$$G_Y(s) := \frac{\mathbb{E}Y^s}{\Gamma(s+1)}$$

is log-concave on $(-1, \infty)$. We have $G_Y(0) = 1$.

By Theorem 4.5 of the source paper, applied at the integer exponent $k = 5$,

$$\mathbb{E}Y^5 \geq \min_{1 \leq m \leq n} \rho(m, 5).$$

Lemma 1 gives $\min_m \rho(m, 5) = \rho(1, 5) = \Gamma(6)$, hence

$$G_Y(5) = \frac{\mathbb{E}Y^5}{\Gamma(6)} \geq 1.$$

For $0 \leq q \leq 5$, log-concavity now yields

$$G_Y(q) \geq G_Y(0)^{1-q/5} G_Y(5)^{q/5} \geq 1.$$

Thus $\mathbb{E}Y^q \geq \Gamma(q+1)$. A final application of Lemma 1 identifies $\Gamma(q+1) = \rho(1, q)$ as $\min_{1 \leq m \leq n} \rho(m, q)$, proving the conjectured inequality in the claimed range. $\square$

5 Verification and novelty notes

The auxiliary script `code/check_q_le_5_endpoint.py` numerically checks the scalar endpoint ratios $R_m(5)$, verifies the finite-grid minimum $\rho(m, q)/\Gamma(q+1) \geq 1$ for $0 \leq q \leq 5$, and records that the naive integer-endpoint interpolation strategy fails just beyond $q = 5$. These computations are not used as proof.

Local index search found no previous packet for arXiv:2512.12254 or this specific $q \leq 5$ consequence. A bounded web/arXiv search on 2026-07-18 found the later adjacent preprint arXiv:2602.03058, *Moments of sums of exponentials, beyond CHS*. Its Theorem 2 proves Schur-concavity of the non-negative moment map for $0 \leq p \leq 4$, which gives the source conjecture in the already-known range $q \leq 4$. The present packet extends the all-dimensional verified range to $q \leq 5$ by a different endpoint argument.

6 Limitations

This is not a full solution of Conjecture 4.8. The interval $q > 5$ remains open here. The same Borell interpolation method cannot simply continue from the integer endpoints 5 and 6: numerically, near $q = 5.35$, the chord between the known integer endpoint lower bounds is below the conjectured minimum of the $\rho(m, q)$ curve.

7 Human review recommendation

This packet should be reviewed as a likely valid partial solution. The most important checks are the invocation of Theorem 4.5 at $k = 5$, the log-concavity of the convolution density for positive coefficients, and the scalar Lemma 1.

References

- [1] S. Brazitikos and C. Pandis, *Sharp inequalities for symmetric polynomials, Hunter's conjecture, and moments of exponential random variables*, arXiv:2512.12254, 2025.
- [2] S. Brazitikos, C. Tang, and T. Tkocz, *Moments of sums of exponentials, beyond CHS*, arXiv:2602.03058, 2026.
- [3] A. Prekopa, Logarithmic concave measures with application to stochastic programming, *Acta Scientiarum Mathematicarum* 32 (1971), 301–316.